pf-qaq — ThGEM DAQ PC: operations overview

As of 7 October 2026

1. Basics

Host	`pf-qaq` (Ubuntu 24.04, kernel 7.0, 56 threads, 251 GB RAM)
Access	`daq@192.168.144.142` on the experiment-site network (from the office, through the VPN). The address is assigned by DHCP and may change.
Digitizer	One VX1730B (SN 66, DPP-PSD) on A3818 optical link 0. Links 1–3 are free; the configuration has templates for more boards.
Experiment name	`X730_ThGEM_Test`. Run numbers continue from the previous ThGEM PC (side3): the last run is 58, **the next run is 59**.

Channel map:

- ch0: LaBr3
- ch1: ThGEM1 trigger. ch2 / ch3 / ch5 / ch4: ThGEM1 XL / XR / YU / YD
- ch6: ThGEM2 trigger. ch7 / ch8 / ch9 / ch10: ThGEM2 XL / XR / YU / YD

2. Data flow and ports

Reader (PSD1) → Merger → three consumers:

Consumer	Output	Port
Recorder	`.delila` files (the primary record)	—
Monitor	Live histograms	8081
root_sink	`.root` files and a live JSROOT display	8090

Service	Port
Operator (web UI and REST API)	**9091**
MongoDB (run history)	27017
mongo-express	8083
InfluxDB	8086
Grafana	3000
eLog (logbook “ThGEM”)	8082

- The start-up script prints “Mongo Express :8082”. That is wrong: 8082 is the eLog.
- The five Docker services restart automatically after a reboot.

root_sink output. One file per run, for example `run0059_0000_X730_ThGEM_Test.root`, containing:

- **hits tree:** every hit. Times are corrected with the per-channel offsets in `config/time_settings_thgem.json`.
- **events tree:** hits grouped around a trigger within ±2000 ns. ThGEM1 is the preferred reference, ThGEM2 the second.

Next to each file, `_eventDef.json`, `_timeSettings.json` and `_hists.json` record which settings produced it. The old `built` tree is no longer written. `tools/root_sink/thgem_events.C` rebuilds the same rows offline from the events tree.

3. Storage

Location	Contents	Used
`/data` (NVMe 3.5 TB, XFS)	Recording (`/data/DELILA_data`) and analysis	2 %
`/archive` (ZFS raidz2, 4 × 20 TB HDD, about 35 TiB usable)	Long-term copy, including the data from side3	1 %

- **Copy:** every 15 minutes, `/data/DELILA_data` is copied to `/archive/DELILA_data`. Nothing is ever deleted from the archive.
- **Snapshot:** a ZFS snapshot of the archive is taken every day.
- **Safety net:** if `/data` goes above 80 %, the oldest recorded files that are confirmed identical in `/archive` are removed from `/data` until it is back at 70 %.
- **Analysis output:** put it outside `/data/DELILA_data` (for example in `/data/analysis`). Otherwise the copy and the safety net treat it as recorded data.

4. Daily operation

- **Starting and stopping the DAQ software.** Normally the DAQ software stays up, and runs are started and stopped from the web UI. Restart it only after a reboot, after a software update, or when a component misbehaves.

  ```
  cd ~/delila-rs
  ./scripts/start_daq.sh      # uses config.toml -> config/config_thgem_pfqaq.toml
  ./scripts/stop_daq.sh       # stops all DAQ components (Docker services stay up)
  ```

  - `start_daq.sh` checks that MongoDB (Docker) is up and starts every component, including root_sink. It then prints the URLs. Logs are in `~/delila-rs/logs/latest/`.
  - **Never run it during a run:** it first kills any DAQ processes that are still running. Stop the run first.
  - Do not add `--no-mongo`. With it, the Operator runs without MongoDB and the run numbers and run history are lost.
  - It can be run from an ssh session as usual; the components keep running after you log out.

- **Runs:** start and stop runs from the web UI at `http://192.168.144.142:9091`. Start does Reset → Configure → Arm → Start in one step, so after a DAQ restart you can press Start directly.
- **Checks while running:**
  - root_sink display (port 8090): XY images, time differences, and `/WriteStatus`, which should read OK.
  - Monitor (port 8081).
  - Grafana (port 3000) for the rates.
- **Check after a run:** `check_events.C` rebuilds the events from the hits tree and compares them with the events tree. It should print `IDENTICAL`:

  ```
  cd /data/DELILA_data
  root -l -b -q \
    "$HOME/delila-rs/tools/root_sink/check_events.C+(\"run0059_0000_X730_ThGEM_Test.root\")"
  ```

- **Tune Up:** while Tune Up is on, root_sink still writes `run0000_*.root` files. They do no harm, but they pile up in `/data/DELILA_data`.